Literature Review

2.1 Review of Existing Solutions

2.1.1 Kaggle Winning Solutions (IEEE-CIS Competition)

Description: The first-place team (including Chris Deotte and Konstantin Yakovlev) combined XGBoost, CatBoost, and LightGBM models. Their key innovation was feature engineering: group-aggregation statistics (mean, standard deviation, counts) and a derived client identifier (`UID`, built from `card1`, `addr1`, and `D1`). Time-ordered GroupKFold validation was used, and predictions were post-processed by averaging them within each client group. The reported final scores were public LB 0.9677 and private LB 0.9459 (ROC-AUC) [1].

What they did well (what we leverage):

- Aggregation-based behavioural features that measure whether a transaction is rare or common for a card or client [1].
- Time-respecting validation, which we adopt to mitigate the leakage risk identified in Part 1 [1].
- Systematic feature selection. The V-column groups were reduced by NaN structure and correlation ($r > 0.75$) [1].

What they lacked (what we avoid):

- The objective was leaderboard ROC-AUC. ROC-AUC can look optimistic under severe imbalance, and no decision threshold or cost model was produced [1].
- Multi-model ensembles with hundreds of engineered features give no per-transaction explanation. Only global feature-importance charts were reported [1].
- Client-level post-processing depends on a dataset-specific identifier reconstruction.

2.1.2 Standard ML and EDA Notebooks

Description: Public notebooks, such as Artgor's "EDA and Models" [2], typically combine data exploration, basic feature engineering, and gradient-boosted models evaluated on the competition metric.

What they did well (what we leverage):

- Visual diagnosis of missingness, class imbalance, and distributions, which supports our Weeks 1–2 EDA phase.
- Reproducible, accessible baselines that show which model families are competitive.

What they lacked (what we avoid):

- Evaluation typically stops at ROC-AUC or accuracy, with no precision/recall trade-off, no cost analysis, and no investigator-facing output.
- No explanation layer and no deployment-oriented component.

2.1.3 Traditional Rule-Based Banking Systems

Description: Rule engines use fixed, human-written conditions (amount, velocity, geography, device) to approve, review, or block transactions [3], [5].

What they do well (what we leverage):

- High transparency. Every alert maps to an explicit rule, which also suits compliance and reporting needs [3], [5].
- Immediate control over known fraud patterns.

What they lack (what we avoid):

- Static thresholds do not learn and adapt slowly to new fraud tactics [3].
- Rule sets that grow and interact can conflict, producing both false negatives and false positives with operational cost [3].
- Excess false declines increase manual review and harm customer experience [5].

2.2 Comparative Analysis

Entries for "Our Proposed System" are **design targets, not validated results**. Consistent with Part 1, results will be reported only after testing.

Criterion	Kaggle Winning Solutions	Standard ML/EDA Notebooks	Rule-Based Banking Systems	**Our Proposed System**
Adaptability to new patterns	High (retrainable, learned features)	Medium (retrainable, limited engineering)	Low (manual rule updates)	High (retrainable ML pipeline)
False-positive handling	Not addressed (no threshold or cost analysis)	Not addressed	Weak (static thresholds, high review load)	Explicit (threshold tuning, decision-cost analysis)
Explainability	Low (ensemble, global importance only)	Low–Medium (feature importance)	High (explicit rules)	Medium–High (SHAP per-transaction; limited by masked features)
Business decision support	None (leaderboard objective)	None	Binary approve/review/block	Prioritised risk score for investigators
Imbalance-aware evaluation	ROC-AUC only	ROC-AUC / accuracy	Not applicable	Precision, Recall, F1, PR-AUC
Deployment practicality	Low (complex, competition-tuned)	Low	High	Prototype only (academic scope)

2.3 Research Gap and Our Contribution

Gap. The three approaches each solve one dimension. Competition solutions maximise predictive power but ignore explanation and decision cost. Notebooks aid understanding but do not produce decisions. Rule engines are transparent but inflexible and costly in false positives. None of the reviewed approaches combines imbalance-aware learning, per-transaction explanation, and an investigator-oriented risk ranking.

Contribution. Our system targets this combination:

1. **Robust ML for imbalance:** gradient-boosted baselines, tested balancing techniques, and time-based validation, evaluated with PR-AUC, Precision, and Recall.
2. **Transparency through XAI:** SHAP-based explanations [6] for individual predictions, so an investigator sees why a transaction was flagged.
3. **Risk scoring for human review:** calibrated probabilities converted into a ranked queue, with thresholds chosen through explicit false-positive and false-negative cost analysis.

Limitations (stated for academic honesty, consistent with Part 1).

- Anonymised features may make SHAP explanations technically valid but hard to translate into business language.
- The dataset is not Egyptian banking data, so no claims about local performance are made.
- Cost parameters will be assumptions, not bank-validated figures.

References

[1] C. McDonald and C. Deotte, “Leveraging Machine Learning to Detect Fraud: Tips to Developing a Winning Kaggle Solution,” NVIDIA Technical Blog, Jan. 2021. https://developer.nvidia.com/blog/leveraging-machine-learning-to-detect-fraud-tips-to-developing-a-winning-kaggle-solution/

[2] Artgor, “EDA and models,” Kaggle Notebook, IEEE-CIS Fraud Detection. https://www.kaggle.com/code/artgor/eda-and-models

[3] Springer chapter on limitations of rule-based fraud detection systems and AI-based alternatives. https://link.springer.com/chapter/10.1007/978-3-030-94590-9_15

[4] IEEE Computational Intelligence Society and Vesta Corporation, “IEEE-CIS Fraud Detection,” Kaggle competition. https://www.kaggle.com/c/ieee-fraud-detection

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[6] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” Advances in Neural Information Processing Systems (NeurIPS), 2017.